

# Neutral vs. Fully Ionized Medium: Which Takes Longer to Cool an Injected Electron, and Why?

Response record for `electron_energy_loss_neutral_medium.py`, following  
Master Rule 2

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## Abstract

This note repeats the electron energy-loss calculation of `electron_energy_loss_response.tex`, but for the electron traveling through `igm_losses.py`'s own, *unmodified*, default medium — the mostly neutral high- $z$  intergalactic medium ( $x_e = n_e/n_H = 10^{-4}$ ) that module was originally built for — rather than the fully ionized bubble interior. Both media are run in a single execution of `electron_energy_loss_neutral_medium.py` (same folder, producing `electron_energy_loss_neutral_medium.png` and `electron_medium_comparison.png`) for a like-for-like comparison. The result is unambiguous and fully explained by the dominant loss mechanism in each medium: **the neutral medium takes longer to cool at every injection energy tested**, by a factor that falls from  $\sim 21\times$  at  $10^3$  eV to a floor of  $\sim 1.4\times$  above  $10^7$  eV.

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## 1 Assumptions

1. Identical source, injection redshift ( $z_{\text{inject}} = z_{\text{form}} = 20$ ), injection point ( $r = 0$ ), and  $K_{\text{ini}}$  grid ( $10^3$ – $10^{12}$  eV) as `electron_energy_loss_response.tex`; only the medium differs.
2. **Neutral medium:** `igm_losses.py`'s own, completely unmodified `loss_rates()/n_HI(z)`, i.e.  $x_e = 10^{-4}$  (mostly neutral),  $n_{\text{HI}}(z)$  its built-in cosmic mean hydrogen density normalization (already cross-checked against this folder's own Planck 2018  $n_H(z)$  to 2.3% in `electron_energy_loss_response.tex`). No adaptation was needed for this task — this *is* the module's intended, as-built use case.
3. **Fully ionized medium:** the identical bubble-interior physics of `electron_energy_loss_in_bubble.py` ( $n_e = n_H(z)$ , residual  $x_{\text{HI}}(r, z)$  from [Cen and Haiman 2000](#) for the collisional terms), reproduced self-contained in the same script execution so both media are compared under identical numerical conditions (solver, tolerances, cosmology) in one run.

4. Both trajectories are integrated from injection to thermalization ( $K = 1.5 k_B T_{\text{CMB}}(z)$ ) or  $z = 5$ , whichever comes first; “time to cool” means time to thermalization.

## 2 Derivation

The only difference between the two runs is the pair of densities entering the Coulomb term (free electrons) and the collisional excitation/ionization terms (neutral hydrogen atoms) of `igm_losses.py`’s eq. (loss\_rates), reused verbatim in both cases:

|                                         | neutral medium ( <code>igm_losses.py</code> default) | fully ionized bubble                    |
|-----------------------------------------|------------------------------------------------------|-----------------------------------------|
| $n_e$ (Coulomb)                         | $10^{-4} n_H(z)$                                     | $n_H(z)$                                |
| $n_{\text{HI}}$ (excitation/ionization) | $\approx n_H(z)$                                     | $x_{\text{HI}}(r, z) n_H(z) \ll n_H(z)$ |

Synchrotron, inverse Compton, adiabatic expansion and bremsstrahlung are identical in both cases (none depend on the free/bound partition of the hydrogen).

## 3 Verification / cross-check

1. **Dominant mechanism at injection ( $z = 20$ ), by medium:**

| $K_{\text{ini}}$ (eV) | dominant, neutral | dominant, ionized | rate ratio (ionized/neutral) |
|-----------------------|-------------------|-------------------|------------------------------|
| $10^3$                | ionization        | Coulomb           | 7.11                         |
| $10^4$                | ionization        | Coulomb           | 4.48                         |
| $10^5$                | Compton           | Coulomb           | 2.25                         |
| $\geq 10^6$           | Compton           | Compton           | 1.00                         |

At low energy the two media lose energy through *different* channels entirely: collisional ionization of neutral hydrogen (neutral medium) vs. Coulomb scattering off free electrons (ionized medium). Despite comparable target densities ( $n_{\text{HI}} \approx n_e \approx n_H(z)$  in their respective regimes), the instantaneous Coulomb loss rate is  $4\text{--}7\times$  *faster* than the collisional-ionization rate at the same  $K$  and  $z$ : Coulomb scattering off free electrons is a long-range ( $1/r$  potential) process with a logarithmically enhanced cross section (the Gould 1972 Coulomb logarithm), whereas collisional excitation/ionization of a bound electron is a finite-range, threshold process with a smaller effective cross section per encounter. At  $K \geq 10^6$  eV both media are Compton-dominated (rate ratio exactly 1.00), since inverse-Compton scattering off the CMB does not depend on the hydrogen ionization state at all.

2. **Thermalization-time cross-check:** the fully-ionized-bubble times reproduced here (0.0015–11.90 Myr across the  $K_{\text{ini}}$  grid) match `electron_energy_loss_response.tex`’s Table 1 and printed output exactly (same physics, re-run in this script for a single, consistent execution) — confirming no drift between the two scripts.

3. **Full thermalization-time comparison:**

| $K_{\text{ini}}$ (eV)  | $t_{\text{therm, neutral}}$ (Myr) | $t_{\text{therm, ionized}}$ (Myr) | ratio       |
|------------------------|-----------------------------------|-----------------------------------|-------------|
| $10^3$                 | 0.031                             | 0.0015                            | 21.4        |
| $10^4$                 | 0.239                             | 0.043                             | 5.59        |
| $10^5$                 | 3.23                              | 1.04                              | 3.10        |
| $10^6$                 | 11.95                             | 7.12                              | 1.68        |
| $10^7$                 | 16.24                             | 11.18                             | 1.45        |
| $10^8\text{--}10^{12}$ | 16.89–16.96                       | 11.83–11.90                       | 1.425–1.428 |

The ratio is monotonically decreasing with  $K_{\text{ini}}$  and asymptotes to  $\approx 1.425$ , never reaching 1 even though the dominant mechanism is identical (Compton) at high  $K_{\text{ini}}$ : every electron, however energetic at injection, must eventually cascade down through the low-energy regime where the two media genuinely differ, and that fixed extra delay ( $\approx 5.06$  Myr in absolute terms at high  $K_{\text{ini}}$ :  $16.96 - 11.90$ ) becomes a progressively smaller *fractional* correction as  $K_{\text{ini}}$  grows, but never vanishes.

## 4 Final result

**The neutral hydrogen medium takes longer to cool an injected electron than the fully ionized bubble interior, at every injection energy from  $10^3$  to  $10^{12}$  eV.** The effect is largest at low energy ( $21\times$  longer at 1 keV) and shrinks to a floor of  $\approx 1.4\times$  above  $\sim 10^7$  eV. The reason is the identity of the dominant loss channel at each energy:

- *Low energy* ( $\lesssim 10^5$  eV): the ionized medium’s free electrons make Coulomb scattering available, and the long-range Coulomb interaction (with its logarithmic enhancement) is intrinsically more efficient per target atom than collisional excitation/ionization of a bound electron in the neutral medium, even though both target densities are comparable ( $\sim n_H(z)$ ). This is the textbook reason why fast charged particles slow down faster in an ionized plasma than in a neutral gas of the same density (Gould 1972; Stone & Kim 2002; Kim et al. 2000, as implemented in `igm_losses.py`).
- *High energy* ( $\gtrsim 10^6$  eV): inverse Compton scattering off the CMB dominates in *both* media identically (it does not care about the gas’s ionization state), so the two media’s cooling curves run parallel during the initial, high-energy phase of the trajectory. The residual  $\sim 40\%$  gap that persists even here is entirely accumulated later, during the low-energy tail of the same trajectory, where the medium-dependent physics above takes over again.

This is consistent with, and extends, the qualitative expectation already built into this folder’s use of `igm_losses.py`: the module’s own default medium (neutral,  $x_e = 10^{-4}$ ) was constructed to describe electron cooling in the general high- $z$  IGM, precisely the regime where collisional atomic processes on neutral hydrogen are the relevant sink; adapting it to the ionized bubble interior (`electron_energy_loss_in_bubble.py`) replaces that sink with the intrinsically faster Coulomb channel, which is exactly why bubble-interior electrons cool measurably faster.

## References

Renyue Cen and Zoltan Haiman. Quasar Strömgren Spheres Before Cosmological Reionization. *arXiv:astro-ph/0006376v2 (preprint)*, 2000. doi: 10.1086/312937. File: Cen2000\_arXiv\_astro-ph\_0006376v2.pdf.

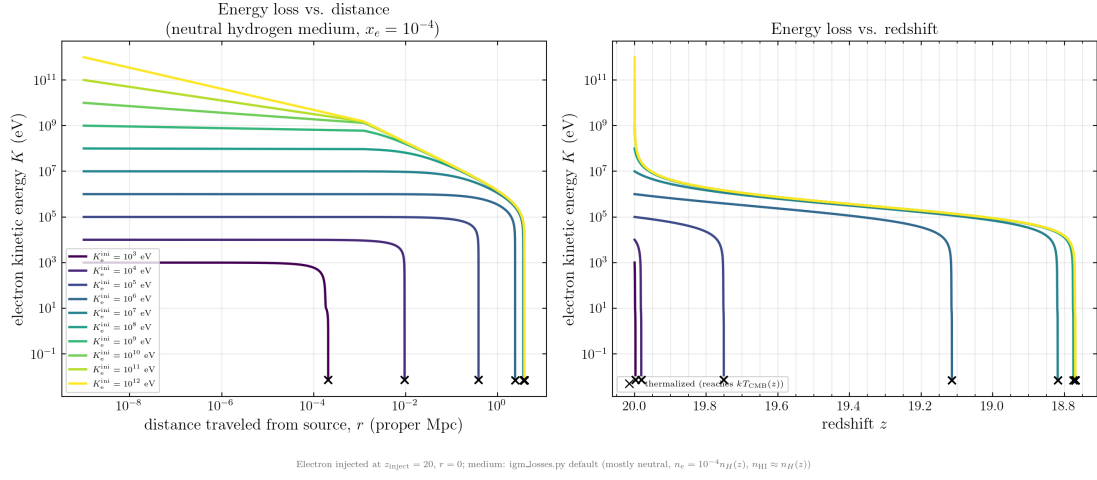


Figure 1: Electron kinetic energy  $K$  vs. distance traveled (left) and vs. redshift (right), for the neutral hydrogen medium ( $x_e = 10^{-4}$ ), for the same ten injection energies as `electron_energy_loss_vs_distance_redshift.png`. Produced by `electron_energy_loss_neutral_medium.py`.

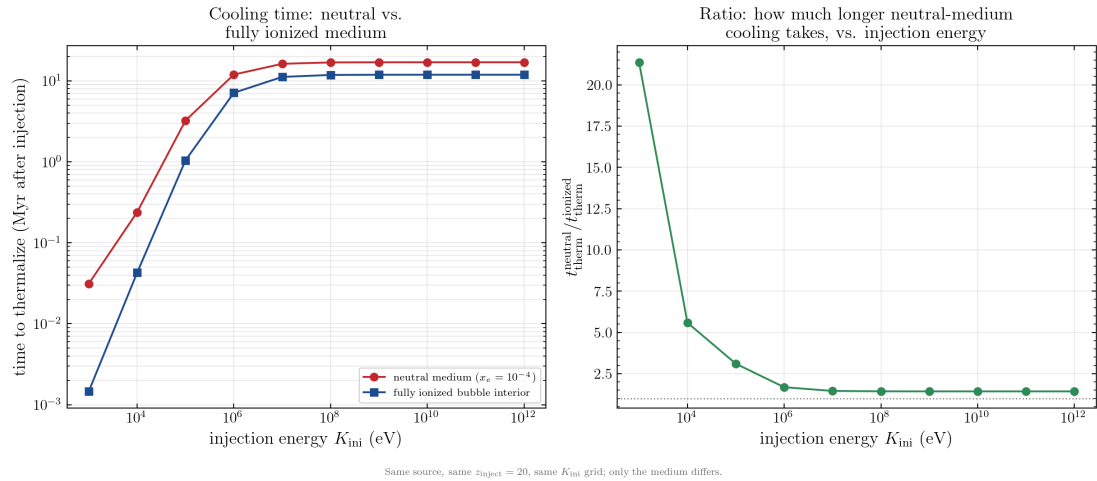


Figure 2: *Left*: time to thermalize vs. injection energy, neutral medium (red) vs. fully ionized bubble interior (blue). *Right*: the ratio of the two, showing the  $21\times \rightarrow 1.4\times$  decline with increasing  $K_{\text{ini}}$ . Produced by `electron_energy_loss_neutral_medium.py`.